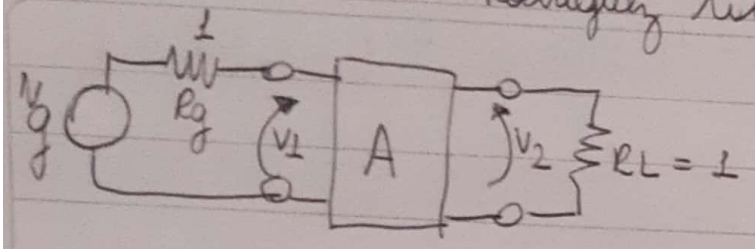
TS 10

Rodríguez Silvera, Martín



LPF Chebyshev 3° orden

LC ; $R_{in} = R_g$;

↳ $|S_{11}|^2 + |S_{21}|^2 = 1$

$$H(\$) = \frac{1}{1+\varepsilon_g^2 \cdot C_n^2(\omega)} = |S_{21}|^2$$

- No especifica rizado → asumo 1 dB.

↳ $\varepsilon_g^2 = 0,258925$

- Rescato de la TS3 la ecuación característica: Cheb N° = 3:

$$\frac{\frac{-1}{16\varepsilon_g^2}}{\$^6 + \frac{3}{4}\$^4 + \frac{9}{16}\$^2 - \frac{1}{16\varepsilon_g^2}} = \frac{-c^2 \cdot b^2}{\$^6 - (c^2+a^2-2b)\$^4 - (2bc^2 - a^2c^2 - b^2)\$^2 - c^2b^2}$$

$$\begin{cases} \frac{1}{16\varepsilon_g^2} = c^2 \cdot b^2 \\ \frac{3}{2} = -a^2 + 2b + c^2 \\ \frac{9}{16} = b^2 - 2bc^2 + a^2c^2 \end{cases}$$

$b^2 = u$; $0 = u^3 - \frac{9}{16}u^2 - \frac{3}{2}\cdot\frac{1}{16\varepsilon_g^2}\cdot u - \frac{1}{16^2\varepsilon_g^4}$

- Resuelvo con $\varepsilon_g^2 = 0,258925$:

$u_1 = 0,98844$; $u_2 = -0,21 + j\,0,11$; $u_3 = -0,21 - 0,11j$

↳ $b = \sqrt{u_1} \cong 0,9942 \cong 1$ → $a = 0,5085 \cong \frac{1}{2}$; $c = 0,4913 \cong \frac{1}{2}$

$$T_{LPF\ N°3\ Cheb}(\$) = \frac{\frac{1}{2}}{\$ + \frac{1}{2}} \cdot \frac{1}{\$^2 + \frac{1}{2}\$ + 1} = \frac{\frac{1}{2}}{\$^3 + \frac{1}{2}\$^2 + 1\$ + \frac{1}{2}\$^2 + \frac{1}{4}\$ + \frac{1}{2}}$$

↳ $= \frac{\frac{1}{2}}{\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2}} = S_{21} = \frac{P(\$)}{Q(\$)}$ ⇒ $S_{11}(\$)\cdot S_{11}(-\$) = \frac{Q(\$)\cdot Q(-\$) - P^2}{Q(\$)\cdot Q(-\$)}$

↳ $|S_{11}|^2 = \frac{(\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2})\cdot(-\$^3 + \$^2 - \frac{5}{4}\$ + \frac{1}{2}) - \frac{1}{4}}{(\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2})\cdot(-\$^3 + \$^2 - \frac{5}{4}\$ + \frac{1}{2})} = \frac{-\$^6 - \frac{9}{16}\$^2}{-\$^6 - \frac{9}{16}\$^2 + \frac{1}{4}}$

Escaneado con CamScanner

Desarrollo... $Q(\$) \cdot Q(-\$) = MQ^2 - NQ^2$

$$|S_{11}|^2 = \frac{\$^6 + \frac{9}{16}\$^2}{\$^6 + \frac{9}{16}\$^2 - \frac{1}{4}} \Rightarrow S_{11}(\$) = \frac{\$\cdot(\$^2 + \frac{3}{4})}{\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2}}$$

$$S_{11}(\$) = \frac{Z_1 - R_{01}}{Z_1 + R_{01}} \Rightarrow Z_1 = R_{01} \cdot \frac{1 + S_{11}(\$)}{1 - S_{11}(\$)} = R_{01} \cdot \frac{V+U}{V-U}, \text{ con } S_{11} = \frac{U}{V};$$

$$Z_1 = R_{01}^{1} \cdot \frac{(\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2}) + (\$(\$^2 + \frac{3}{4}))}{(\$^3 + \$^2 + \frac{5}{4}\$ + \frac{1}{2}) - (\$^3 + \frac{3}{4}\$)} = \boxed{\frac{2\$^3 + \$^2 + 2\$ + (1/2)}{\$^2 + \frac{1}{2}\$ + \frac{1}{2}}}$$

- Síntesis de izquierda - derecha Cauer ▷ : INF.

$2\$^3 + \$^2 + 2\$ + \frac{1}{2} \;\big|\; \$^2 + \frac{1}{2}\$ + \frac{1}{2}$

$-2\$^3 + \$^2 + \$ + 0 \;\big|\; 2\$$ → $2\$$

$0 + 0 + \$ + \frac{1}{2}$

$\$^2 + \frac{1}{2}\$ + \frac{1}{2} \;\big|\; \$$

$-\$^2 + \frac{1}{2}\$ + 0$ → $\$$

$0 + 0 + \frac{1}{2}$

$\$ + \frac{1}{2} \;\big|\; 2\$$

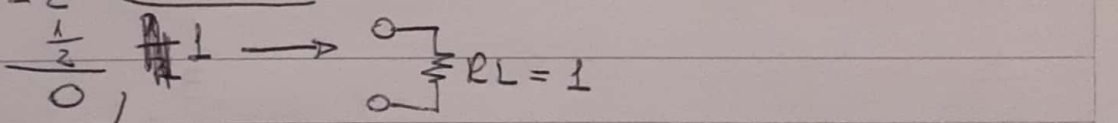
$-\$ + 0$ → $2\$$

$0 + \frac{1}{2}$

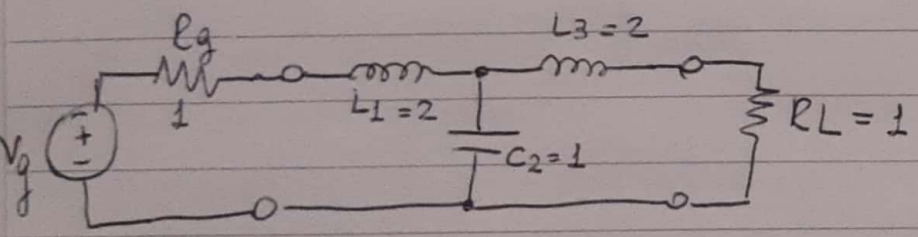
$\frac{1}{2} \;\big|\; 1$

$-\frac{1}{2}$ → 1

0



- Circuito sintetizado:



Escaneado con CamScanner